

Chapter 42

Missing verbal structure in Bosnian-Croatian-Serbian agent nominals

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Based on morpho-phonological and syntactic evidence from BCS agent nominals, I argue that elements which are traditionally analyzed as verbal in Slavic ('verbal' theme vowels, secondary imperfectivizers, lexical prefixes) are not verbal at all. Illustrating with data from root-derived versus deadjectival agent nominals, I first show that root-conditioned allomorphy and accent placement in BCS are limited to the first spell-out domain, which may include only one categorizer. I then show that agent nominals containing morphology analyzed as verbal behave for these morpho-phonological processes like root-derived nouns. I argue there is no syntactic evidence for the presence of verbal structure in BCS agent nominals, as well as that the available semantic evidence (the presence of event/agent entailments) should not compel us to assume such structure either. Finally, I offer a tentative alternative analysis for the identity of the 'verbal' morphemes in question, treating BCS agent nominals as root-root compounds.

Keywords: agent nominal, verbal structure, theme vowel, secondary imperfective, root-root compound, event entailment, Bosnian-Croatian-Serbian

1 Introduction

In this paper, I argue that morphemes traditionally taken to be part of the verbal extended projection are not verbal at all in Bosnian-Croatian-Serbian (BCS) agent nominals. I will be concerned primarily with a sample of BCS agent nominals illustrated in (1)-(3), which contain what is traditionally analyzed as verbal morphology across Slavic languages.

- | | | | |
|-----|---|--|---|
| (1) | a. prouč-av-a-telj
study-AV-TH-N
'researcher' | b. pozn-av-a-telj
know-AV-TH-N
'expert' | c. reš-av-a-telj
solve-AV-TH-N
'solver' |
| (2) | a. predsed-av-a-ač
chair-AV-TH-N
'chair' | b. pred-av-a-ač
lecture-AV-TH-N
'lecturer' | c. ugnjet-av-a-ač
oppress-AV-TH-N
'oppressor' |
| (3) | a. prod-av-a-ac
sell-AV-TH-N
'seller' | b. dar-o-d-av-a-ac
gift-L-give-AV-TH-N
'giftgiver' | c. posl-o-d-av-a-ac
job-L-give-AV-TH-N
'employer' |

Specifically, when *-av* (or one of the morpheme's other allomorphs) appears on the corresponding verbs (4), it is traditionally analyzed as a 'secondary imperfectivizer', given its role in producing atelic/imperfective verbal stems from the telic/perfective ones in (5) (e.g., Schuyt 1990; see Łazarczyk 2009 and Tatévoso 2015 for a discussion of the precise role of this morpheme in the verbal domain). This is an entirely productive process in BCS. Note also that the agent nominals in (1)-(3) contain the same theme vowels as the imperfective verbs in (4). These theme vowels have been claimed to be exponents of the verbal categorizing head (*v*); see, for example, Svenonius 2004a, Čaha & Ziková 2016, Biskup 2019, Milosavljević & Arsenijević 2022, Bešlin 2023.¹

- | | | | |
|-----|---|--|---|
| (4) | a. prouč-av-a-ti
study-AV-TH-INF
'be researching' | b. predsed-av-a-ti
chair-AV-TH-INF
'be chairing' | c. prod-av-a-ti
sell-AV-TH-INF
'be selling' |
| (5) | a. prouč-i-ti
study-TH-INF
'research' | b. predsed-a-ti
chair-TH-INF
'chair' | c. prod-a-ti
sell-TH-INF
'sell' |

Let me note at the outset that the presence of the theme vowel can only be detected at the surface in the agent nominals in (1), because the nominal suffixes

¹Many of the nominals I discuss also contain so-called *lexical prefixes*, which have been studied extensively in the verbal domain (e.g., Svenonius 2004c), cf. (ia-b). I will ignore these prefixes in the glosses for the time-being and return to them in section 4.3.

- | | | |
|-----|---|---|
| (i) | a. pro-uč-i-ti
LP-learn-TH-INF
'research' | b. uč-i-ti
learn-TH-INF
'be learning' |
|-----|---|---|

in (2)-(3) begin with a vowel. In Slavic, this kind of hiatus is frequently resolved by deleting the first (leftmost) vowel, here the theme vowel (Jakobson 1948). The deletion is a phonological process, so the theme vowel is present at the form interface (for example, at Vocabulary Insertion), which will be of interest to us here. While I will continue to represent the theme vowel throughout, it is not crucial for any of my arguments that it be present on all relevant agent nominals. It is sufficient that it be present on at least some of them, which is apparent from the surface forms. The claim is that, for those agent nominals that contain the theme vowel or the ‘secondary imperfectivizer’, these morphemes are not verbal.

The nominals in (1)-(3) contain different *n*-allomorphs and accent in these agent nominals can surface on the *n*-head. Building on Bešlin 2025b, to appear, I will show that BCS root-conditioned allomorphy and accent placement are limited to the first spellout domain, which includes only one categorizing morpheme. Given the observed patterns of allomorphy and accent placement in the agent nominals in (1)-(3), I will argue that the ‘verbal’ elements we see inside them are not verbal at all and raise the possibility that these elements are not exponents of verbal functional heads even when they are found on verbs.

The paper is organized as follows. Section 2 provides some necessary background on Distributed Morphology (DM), cyclic domains, the role of categorizers, and allomorphy. Illustrating with data from root-derived versus deadjectival agent nominals, section 3 shows that root-sensitive allomorphy and accent placement in BCS are confined to the first spellout domain, centered around the first-merged categorizer. This behavior is shown to follow from a DM conception of cyclic domains. Section 4 returns to the ‘deverbal’ agent nominals discussed in this section, arguing that they pattern with root-derived nouns (and root-root compounds) for purposes of allomorphy and accent placement. It then provides an alternative analysis of the ‘verbal’ morphemes under discussion, arguing that they are not, in fact, part of the verbal extended projection. It is proposed that the morpheme AV is a root, that theme vowels are elements that attach to (certain) roots more generally, and that the so-called ‘verbal lexical prefixes’ cannot be verbal, since they appear in contexts in which a deverbal analysis is very dubious. This section also provides a structural analysis of BCS agent nominals as root-root compounds. Finally, this section challenges existing meaning-based arguments for assuming verbal structure in agent nominals. Section 6 concludes.

2 Theoretical background

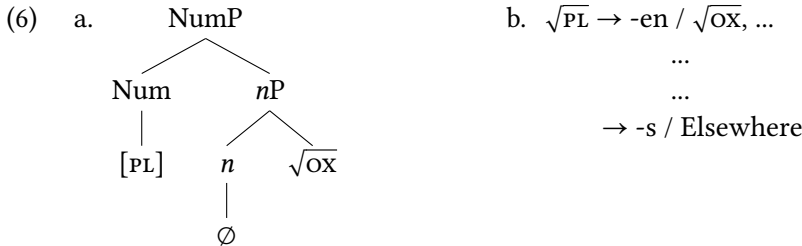
In this section, I briefly present relevant aspects of the framework I couch my analysis in, Distributed Morphology (DM). There are several aspects of DM which

make it particularly opportune for capturing the phenomena I address in this paper, namely syntactically conditioned locality effects at the form interface.

First, DM is a piece-based approach to morphology. The pieces (syntactic atoms) include roots and functional heads. Syntactic categoryhood is a derived notion: A categorial roots obtain their category in syntactic derivations with the obligatory addition of categorizers, *n*, *v* and *a* (for nouns, verbs, and adjectives).

DM is also a realizational framework. Words are built up syntactically out of abstract morphemes that receive form and meaning at the relevant interfaces. Focusing on form (with similar issues arising at the meaning interface, see [Marantz 2013](#), [Myler 2014](#), [Wood 2023](#)), this allows us to capture the fact that the form of one morpheme may be determined by the identity or morphosyntactic features of another morpheme in its environment.

For example, the plural morpheme will have the form *-en* if it is in the context of (among others) the root $\sqrt{\text{OX}}$ (6). Different allomorphs of a single morpheme are thought to be in competition with each other, regulated by the *Subset Principle* (also known as the *Elsewhere Condition*). Vocabulary Insertion (VI) lists such as (6b) are consulted starting with the most specific (i.e., most contextually restricted) Vocabulary Items. If the context for the insertion of the more specific allomorph(s) is not met, the elsewhere allomorph is inserted.



Morphemes cannot influence each other's form across unbounded distances, however. One way in which these interactions have been argued to be constrained is cyclic spell-out. Spell-out (transfer to the interfaces) happens cyclically, at certain points of the derivation, with the categorizers (*v*, *n*, *a*) the relevant cyclic heads. The schema in (7) provides a description of the spell-out mechanism.

- (7) SCHEMATIZATION OF CYCLIC DOMAINS ([Embick 2014](#)):
- a. Cyclic *y* merged in $[y [X [Y [x \sqrt{\text{ROOT}} \dots]]]]$
 - b. Cyclic domain centered on $x = [X [Y [x \sqrt{\text{ROOT}}]]]$ sent to interfaces

The intended outcomes of (7) are as follows: The root is accessible to the first cyclic head x and any non-cyclic heads (X, Y) in x 's extended projection (as in *ox-oxen* above). Furthermore, the root and the second cyclic head y are not visible to each other (qua morphemes) because they are in separate spell-out domains.

Embick (2014) furthermore notes that what is sent to the interfaces at one cycle is not inaccessible at the next cycle in its entirety. Specifically, while the root and cyclic x (along with any non-cyclic heads in x 's extended projection) are spelled-out in the same cycle, only the complement of x (i.e., the root) becomes inaccessible at the cycle y is spelled out (8). On the other hand, x and any functional heads above it are still visible to y in the cycle y undergoes VI. We will see the effects of the Activity corollary in action in section 3.

(8) ACTIVITY COROLLARY (Embick 2014):

In $[[\dots x] \dots y]$, x and y cyclic, the complement of x is not active in the PF cycle in which y is spelled out.

Cyclic spell-out is thought to explain many patterns of (im)possible morpho-phonological interactions, including (im)possible allomorphy. A prominent early example discussed in Marantz 1997 is that of two kinds of nominals in English, what Chomsky 1970 calls derived nominals (9) versus gerundive nominals (10). There are a number of different exponents of n in English, as exemplified in (9). The different allomorphs are not interchangeable; the choice of n is based on the identity of the root it combines with. On the other hand, gerundive nominals show uniform nominal morphology across roots.

(9) *marri-age*, *grow-th*, *remov-al*, *free-dom*, *divers-ity*, *strateg-y*, ...

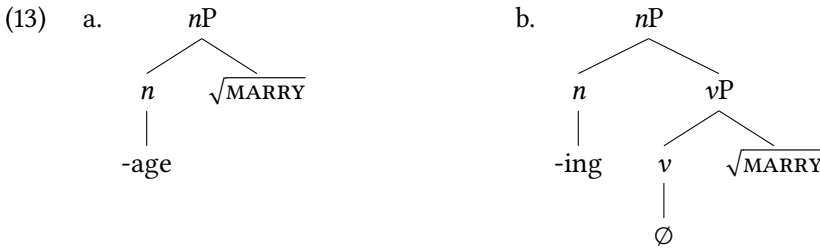
(10) *marry-ing*, *grow-ing*, *remov-ing*, *free-ing*, *divers-ify-ing*, *strateg-iz-ing*,...

Why would this be? The elements in (9)/(10) are all nominal; they appear in clausal positions that are occupied only by nominals, and can be accompanied by nominal possessors (11)/(12). However, gerunds exhibit syntactico-semantic (and sometimes morphological) properties that suggest they contain some verbal structure. This includes the following properties: (i) gerunds have accusative objects, despite the fact that simple nouns in English require *of*-complements (e.g., *a student of physics*); (ii) gerunds are modified by adverbs, not adjectives; (iii) gerunds have an eventive interpretation, which has been associated with the presence of verbal structure; (iv) some gerunds include overt *vs*, for example *-ify* in *diversifying*.

(11) We were surprised by [their thorough diversity of opinions].

(12) We were surprised by [their diversifying their investments thoroughly].

Given all this, we may conclude that gerunds, but not derived nominals, contain verbal structure. I give a schematic structural representation of the two types of nominals in (13).² The structural differences between the two can also explain the availability of root-conditioned *n*-allomorphy in (9) but not (10). Given the cyclicity schema in (7), the root is accessible when *n* undergoes spell-out in (13a) since it is the first-merged categorizer. The root is not accessible when *n* undergoes spellout in (13b) and therefore cannot influence the choice of *n*-allomorph in this configuration.



Intervention by a categorial head has the same effect on allomorphy and accent placement in BCS, as I show in detail in the next section. The first (root-centered) spell-out domain includes the first categorizer and any functional heads in its extended projection, but not the second categorizer.

3 Constraints on allomorphy and accent placement in BCS

In this section, I report and build on data from Bešlin 2025b, to appear demonstrating that BCS categorizers *a* and *n* are cyclic heads. Being cyclic heads, *a* and *n* impose locality boundaries for morpho-phonological processes. More specifically, I contrast the behavior of root-derived and deadjectival (agentive) nouns (14), showing that the presence of *a* in deadjectival nouns prevents *n* from participating in first-phase morphophonological processes: root-conditioned allomorphy and accent placement.³

²Some existing work treats derived nominals with a particular interpretation as having a more elaborate internal structure (e.g., Alexiadou 2001). See section 5 for arguments against using certain types of semantic evidence as evidence for the presence of syntactic structure.

³As we will see, accent placement in BCS is limited to the first spellout domain. This is a property of the phonological system of BCS (as well as e.g., Turkish and Cupeño, Newell 2008). Other languages may have their lexical stress ‘recalculated’ at every cycle within a phonological word

- (14) a. [[ROOT] *n*] b. [[[ROOT] *a*] *n*]

I show that the intervention effect of *a* cannot be explained in simple (linear or structural) adjacency terms, by contrasting its behavior with the behavior of non-categorizers, including negation, comparatives, and roots in root-root nominal compounds. In section 4, I show that the ‘deverbal’ *-av(a)* agent nominals pattern with root-derived nouns and root-root nominal compounds, rather than with nouns whose derivation involves recategorization. This suggests that the structure inside *-av(a)* nouns is not verbal.

3.1 Allomorphy in root-derived versus deadjectival agent nominals

As illustrated in (15), the broadly agentive *n*-suffixes in BCS are at least *-telj*, *-ar*, *-aš*, *-er*, *-an*, *-a*, *-(a)c*, *-ač*, *-ic(a)*, *-ik*, *-ist(a)* and *-džij(a)*. Root-derived nouns may take any of the *n*-allomorphs on offer; the choice of *n* is determined by the root (so-called ‘lexically-conditioned allomorphy’).

- (15) a. uči-*telj* 'teacher' g. pis-*ac* 'writer'
 b. kormil-*ar* 'helmsman' h. voz-*ač* 'driver'
 c. batin-*aš* 'beater' i. izdaj-*ica* 'traitor'
 d. poz-*er* 'poser' j. proza-*ik* 'prose writer'
 e. blebet-*an* 'chatterbox' k. šah-*ista* 'chess player'
 f. dangub-*a* 'loiterer' l. bureg-*džija* 'börek maker'

A root may determine the choice of n as in (15) only if n is the first-merged categorizer. If a intervenes between the root and n (ROOT- a - n), the root can no longer determine the form of n (16)-(19).⁴ Only a can now influence the form of n , which is uniform regardless of the root in question (either due to a -conditioned allomorphy or the application of the *Elsewhere principle*). Note that the forms in (16)-(19) are not possible without the presence of the a -head (e.g., **plaš-ac*), confirming that it is not the root that conditions the n -allomorph in these forms.

(e.g., Armenian, [Dolatian 2020](#) and Farsi, [Amini 1997](#)). There are different ways to implement the observed cross-linguistic variation depending on one’s assumptions about the phonology. The implementation of this particular issue is irrelevant for my purposes here.

⁴In section 4.1, I will tentatively propose that the morpheme *av* on agent nominals may be a root. If the *-av* we see in adjectives like (16) is that same *av* (as argued for in Quaglia et al. 2022), then (16) may involve (null) adjectivization of a root-root compound, followed by nominalization. As we will see in section 3.2, the accent pattern on these deadjectival nominals is the same as on the corresponding adjectives (and different than on root nominals with the same *n*-head).

The behavior of these deadjectival nouns is fully in line with the view of cyclic domains presented in (7) and the Activity corollary in (8).

- | | | |
|------|---|---|
| (16) | a. prlj- <i>av-ac</i> ‘dirty one’ | d. peg- <i>av-ac</i> ‘freckled one’ |
| | b. mrš- <i>av-ac</i> ‘skinny one’ | e. prg- <i>av-ac</i> ‘grumpy one’ |
| | c. mut- <i>av-ac</i> ‘mute one’ | f. hvalis- <i>av-ac</i> ‘boastful one’ |
| (17) | a. plaš- <i>ljiv-ac</i> ‘scared one’ | d. povod- <i>ljiv-ac</i> ‘gullible one’ |
| | b. smrd- <i>ljiv-ac</i> ‘stinky one’ | e. var- <i>ljiv-ac</i> ‘cheating one’ |
| | c. grab- <i>ljiv-ac</i> ‘predatory one’ | f. vaš- <i>ljiv-ac</i> ‘lousy one’ |
| (18) | a. hajduk- <i>ov-ac</i> ‘H. supporter’ | d. sps- <i>ov-ac</i> ‘SPS member’ |
| | b. dinam- <i>ov-ac</i> ‘D. supporter’ | e. nobel- <i>ov-ac</i> ‘Nobel winner’ |
| | c. maček- <i>ov-ac</i> ‘M. follower’ | f. oskar- <i>ov-ac</i> ‘Oscar winner’ |
| (19) | a. smrt- <i>n-ik</i> ‘mortal one’ | d. bestid- <i>n-ik</i> ‘shameless one’ |
| | b. put- <i>n-ik</i> ‘traveler’ | e. duž- <i>n-ik</i> ‘debtor’ |
| | c. boles- <i>n-ik</i> ‘sick one’ | f. gubit- <i>n-ik</i> ‘loser’ |

The locality effect is best observed when the same root can produce both a root-derived noun and a deadjectival noun, as in (20); cf. **gubit-n-aš*, **gubit-ik*.⁵ The two nouns in (20) share the same root and have the same meaning, yet they contain different exponents of *n* due to the presence of *a* in (20b).⁶

⁵A reviewer asks why *gubit* in (20) is not decomposed into *gub-i-t* lose-TH-INF, given the existence of the root *gub* in BCS, for example *dan-gub-a* ‘loiterer’ (lit. *day-los-er*). The claim that *gubit* is a root is not necessary; one could make the point by claiming the presence of *a* blocks *n* from accessing information about the verbal stem. However, if *gubitaš* were derived by adding *-aš* to an infinitival stem, we would expect other infinitival stems to be able to undergo such a derivation. While the process need not necessarily be productive to conclude this, I was not able to find *any* other examples of this kind. This lends some support to the idea that *-aš* is simply merged to a root here (as in many other uncontroversial cases). Moreover, the accent on *gubitaš* surfaces on *-aš*. As we will see in the following section, this suggests that *-aš* is in the first spell-out domain and not merged to a verbal stem. Hence, I treat *gub* and *gubit* as two distinct roots in BCS.

⁶In many other examples, the root-derived vs. deadjectival nouns have different meanings, but they demonstrate the same point for the purposes of allomorphy, for example *put-ar* ‘road worker’ vs. *put-n-ik* ‘traveller’ (cf. **put-n-ar*, **put-ik*).

- (20) a. gubit-*aš*
lose-N
'loser'
- b. gubit-*n-ik*
lose-A-N
'loser'

In other words, even though the root $\sqrt{\text{gubit}}$ clearly picks out the nominalizer *-aš*, it can no longer do so if an adjectivizer intervenes between the two. This can be accounted for if BCS *a* and *n* are cyclic heads. In a [[[ROOT] *a*] *n*] configuration, the root and *n* are in separate spellout domains, hence the root (*qua* morpheme) can no longer be identified when *n* undergoes VI and can therefore not influence the choice of *n*-allomorph.

3.2 Accent placement in root-derived vs. deadjectival agent nominals

Exponents of BCS morphemes are idiosyncratically marked or unmarked for accent, the phonetic correlate of which is pitch prominence (Inkelas & Zec 1988). I mark pitch prominence with an acute accent in the following prose.⁷ Bešlin 2025b shows that there are constraints on the realization of accent in polymorphic words. Specifically, accent in BCS is realized on the structurally highest underlyingly accent-marked element in the first spellout domain, where the first spellout domain is understood as in (7). For example, the nominalizer *-(á)c* is underlyingly accent-marked, but it only realizes that accent if it is in the configuration [[[ROOT] *n*] (21), and not, for example, in [[[ROOT] *a*] *n*] (22). I mark the position of the accent with an acute symbol throughout (á).

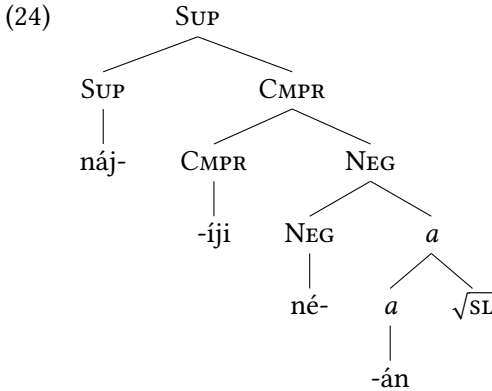
- (21) a. pis → pis-ác
 $\sqrt{\text{write}}$ 'writer'
- b. alžír → alžír-ác
 $\sqrt{\text{algeria}}$ 'Algerian(N)'
- (22) a. pŕlj-av → pŕlj-av-ac
'dirty' 'dirty one'
- b. smrd-ljív → smrd-ljív-ac
'stinky' 'stinky one'

Also in accordance with the cyclicity schema in (7), if the extended projection of the first categorizer contains non-cyclic heads and their exponents are accented, the accent surfaces on (the structurally highest one of) them. These non-cyclic heads include, for example, comparative and superlative degree for adjectives, diminutive for nouns, and negation for both nouns and adjectives. A particularly revealing example is that of adjectives whose derivation includes all

⁷See Bešlin 2025b for an overview of Inkelas & Zec's (1988) influential analysis of the BCS accentual system. This work (p. 177-8) also contains a discussion of the terms *accent*, *stress*, and *morphological tone*.

of the above-mentioned elements. Illustrating with *slan* ‘salty’ (23a), notice that the accent surfaces on the exponent of *a*, but it shifts to the structurally higher negation in the negated form in (23b). The shift occurs again with the addition of the comparative morpheme (23c) and the superlative (23d), which is built overtly on the comparative. The structure of (23) is given in (24). Notice also that the accent repeatedly shifts to the structurally higher (underlyingly accented) element regardless of that element’s status as a prefix or suffix, that is, independently of its linear position to the left or right of the root.

- (23) a. *sl-án*
salt-A
‘salty’
- b. *né-sl-an*
NEG-salt-A
‘unsalty’
- c. *ne-sl-an-íji*
NEG-cold-A-CMPR
‘unsaltier’
- d. *náj-ne-sl-an-iji*
SUP-NEG-salt-A-CMPR
‘most unsalty’



It is possible for some syntactic composition to occur before categorization takes place. Of specific interest here are root-root (nominal) compounds (25). Root-root compounds in BCS frequently involve visibly bound roots which cannot appear as words independently. For example, neither *kras* ‘beauty’ nor *pis* ‘write’ in (25a) can be used without the addition of overt categorizers or other elements. Root-root compounds also include a L(inker) and contain one accent. They behave as expected under the cyclicity schema in (7): Since the roots are merged before any categorization takes place, the accent placement is still ‘frozen’ only in the spellout domain of the first categorizer. In other words, the accent can still surface on *n* if *n* is overt and has an underlying accent (25) or on a suffix in *n*’s extended projection, as illustrated with the diminutive suffix in (26a); cf. the accent

in the base form (26b). What is important is only that these roots are combined before any categorization takes place. In contrast, compounds that involve two categorized elements (e.g., two nouns) are not connected via a *l*(inker) and each element has an accent of its own (27). The behavior of root-root compounds will become important immediately in the following section; we will see that ‘deverbal’ agent nominals in BCS behave for the purposes of accent placement exactly like root-root nominal compounds. I discuss the structure of AV-containing agent nominals as root-root compounds in section 4.5.

- | | | |
|------|--|--|
| (25) | a. kras-o-pis-ác
beauty-l-write-N
‘calligrapher’ | b. pad-o-bran-ác
fall-l-defend-N
‘parachuter’ |
| (26) | a. pad-o-bran-čić
fall-l-defend-DIM
‘little parachute’ | b. pád-o-bran-Ø
fall-l-defend-N
‘parachute’ |
| (27) | a. nadžák bába
pickaxe.N grandma.N
‘belligerent person’ | b. rák rána
cancer.N wound.N
‘sore subject’ |

In this section, I showed that root-conditioned allomorphy and accent placement in BCS are limited to the first spellout domain, which includes exactly one categorizer. In the following section, we will see that BCS ‘deverbal’ agent nominals pattern with respect to these phenomena as though they only contain one categorizer (namely, *n*). This will lead me to argue that the ‘verbal’ elements inside these nominals are not verbal after all.

4 Back to ‘deverbal’ agent nominals

As I showed in section 1, there are different *n*-allomorphs in ‘deverbal’ agent nominals (28)–(30). The material intervening between the root and *n* in these nominals is the same: *-av* and the theme vowel *-a*. There are no syntactic differences that correlate with the presence of different *ns* (e.g., differences in argument structure). There are also no discernible semantic or phonological factors that condition the allomorphy on *n*. I therefore conclude that the *n*-allomorphs in (28)–(30) are conditioned by the identity of the respective roots (what is sometimes called ‘lexically-conditioned allomorphy’). This is very surprising if the material intervening between the root and *n* is verbal. Given the cyclicity schema in (7) and

what we have seen with root-derived versus deadjectival nouns in BCS, root-conditioned allomorphy is restricted to the spellout domain of the first-merged categorizer.

- | | | | |
|------|---|--|---|
| (28) | a. prouč-av-á-telj
study-AV-TH-N
'researcher' | b. pozn-av-á-telj
know-AV-TH-N
'expert' | c. reš-av-á-telj
solve-AV-TH-N
'solver' |
| (29) | a. predsed-av-a-áč
chair-AV-TH-N
'chair' | b. pred-av-a-áč
lecture-AV-TH-N
'lecturer' | c. ugnjet-av-a-áč
oppress-AV-TH-N
'oppressor' |
| (30) | a. prod-av-a-ác
sell-AV-TH-N
'seller' | b. dar-o-d-av-a-ác
gift-L-give-AV-TH-N
'giftgiver' | c. posl-o-d-av-a-ác
job-L-give-AV-TH-N
'employer' |

Furthermore, as also shown in (28)-(30), accent in these 'deverbal' agent nominals can surface on the *n*-exponents that underlyingly have it (-áč and -ác). Recall, accent can only surface in the first spellout domain, which includes one categorizer. Hence, not only allomorphy but also accent placement patterns suggest that the *n* in BCS 'deverbal' agent nominals is the first-merged categorizer. A corollary of this conclusion is that the 'verbal' morphology inside these nominals is not verbal at all.

4.1 The morpheme AV

As we have seen, the morpheme AV, expounded most frequently as *-av* or *-iv*, appears productively in so-called secondary imperfective verbs and signals a shift in aspect (31)-(32).

- | | | |
|------|---|---|
| (31) | a. prouč-i-ti
study-TH-INF
'research' | b. prouč-av-a-ti
study-AV-TH-INF
'be researching' |
| (32) | a. zatašk-a-ti
cover_up-TH-INF
'cover up' | b. zatašk-iv-a-ti
cover_up-IV-TH-INF
'be covering up' |

The morpheme is also found on *some* agent nominals (33), though not all (34a): The nominals in (33) appear to be related to the atelic/imperfective verbs in (31b)/(32b), while the nominal in (34a) appears to be related to the telic/perfective verb in the aspectual pair (34b-c).

- (33) a. prouč-av-a-telj
 study-AV-TH-N
 ‘researcher’

 b. zatašk-iv-a-ač
 coverup-IV-TH-N
 ‘cover up agent’
- (34) a. uruč-i-telj
 serve-TH-N
 ‘process server’

 b. uruč-i-ti
 serve--IV-TH-INF
 ‘serve’

 c. uruč-iv-a-ti
 unload-IV-TH-INF
 ‘be serving’

When present, the morpheme AV is in the same position as on verbs, namely immediately adjacent to the root. However, there are a number of reasons why associating this morpheme with aspectual features is problematic. In the verbal domain, evidence has been mounting in favor of the view which dissociates traditional imperfectivizing affixes from grammatical aspect (e.g., Klein 1995, Łazarczyk 2010, Tatevosov 2015; see Arsenijević, Milosavljević & Simonović 2025: section 2.2.2. for a detailed overview). The majority view has been that the morpheme the morpheme AV is an atelicizing/eventizing element, or that it otherwise manipulates the event content of the predicate.

In this section, I focus on the behavior of the morpheme AV outside the verbal domain. The contexts I consider point to the conclusion that AV in these contexts does not encode (lexical or grammatical) aspect or a specific event-structure-related property. First, nominals with or without the morpheme AV do not denote agents of (aspectually) different kinds of events. Specifically, the nominals in (33) do not necessarily denote agents of atelic/imperfective events, nor do the nominals in (34a) necessarily denote agents of telic/perfective events.

Although most agent nominals are deterministic in terms of whether they have AV (33) or not (34a), there are also a few cases where the agent nominal can be derived with or without the morpheme, with no apparent difference in meaning (35). Crucially for us here, there are no aspectual or event-structural differences in the meanings of the two nominals, as would be expected if AV were an aspectual or event-related morpheme.

- (35) a. ponud^j-i-ač
 offer-TH-N
 ‘contractor’
 b. ponud^j-iv-a-ač
 offer-IV-TH-N
 ‘contractor’

Moreover, the morpheme AV can appear in the context of so-called verbs of creation (37). As noted in Embick 2004, verbs of creation (*build, make, create...*) are incompatible with predicates whose meaning involves event entailments, as shown by the contrast in (36); see also Kratzer 2000 for a similar point. Specifically, the resultative participle *opened* denotes a result state of a prior event, and

this eventive semantics clashes with the verb of creation. The purely stative *open* shows no such clash.

(36) This door was built open/*opened.

Importantly for our purposes, BCS participles which include the morpheme *av* can readily appear in the context of a verb of creation. In (37), the creation of the model is achieved with a 3D printer, so the observable state of being destroyed or crumpled does not necessitate an event of destroying/crumpling. The participle denotes a simple state and involves no event entailments. If the morpheme *av* were an aspectual or event-related morpheme, its denotation would necessarily involve event modification. This should hold regardless of whether *av* is a marker of lexical/grammatical aspect or an event-structure marker. Given that there is no clash with verbs of creation, we see that the morpheme *av* appears in contexts where it clearly does not contribute an event-related meaning component.

(37) 3D štampač je pokvaren pa je maketa izašla
 3D printer is broken so is model came_out
 iz-u-nišť-av-a-n-a / is-pre-sav-ij-a-n-a.
 SP-LP-destroy-AV-TH-PTCP-F.SG SP-LP-bend-IJ-TH-PTCP-F.SG
 ‘The 3D printer is broken so the model came out destroyed/crumpled.’

If *av* is not an aspectual morpheme, what is it then? Quaglia et al. 2022 point out that *-av* and *-iv* do not only appear in the derivation of verbs, but also in the derivation of seemingly simple nouns and adjectives. Based on this, they argue that *av* is a (bound) root (see Lowenstamm 2010 for the proposal that English derivational affixes are roots and Simonović 2022, Simonović et al. 2025 for a Slavic perspective).

(38)	a. ruk-av-Ø arm-AV-N.M.SG.NOM ‘sleeve’	c. maz-iv-o daub-IV-N.NEUT.SG.NOM ‘grease’
	b. bles-av-Ø silly-AV-A.M.SG.NOM ‘silly’	d. jez-iv-o shudder-IV-A.NEUT.SG.NOM ‘creepy’

Recall I have also shown that *av*-containing agent nominals behave exactly the same as root-root nominal compounds for purposes of allomorphy and accent placement. Specifically, the exponent of *n* in both types of nouns behaves as

though it is the first categorizer. I therefore tentatively conclude with Quaglia et al. 2022 that AV is a root.⁸

4.2 Theme vowels

As I mentioned in section 1, the agent nominals discussed in this paper have the same theme vowels as the corresponding verbs derived from the same stems (39)–(40). On verbs, theme vowels have been argued to be exponents of *v* (Svenonius 2004a, Caha & Ziková 2016, Biskup 2019, Milosavljević & Arsenijević 2022, Bešlin 2023). All else equal, the theme vowels on agent nominals should then also be treated as verbal.

- | | | |
|------|---|--|
| (39) | a. uruč-i-ti
serve--IV-TH-INF
‘serve’ | b. uruč-i-telj
serve-TH-N
‘process server’ |
| (40) | a. osigura-av-a-ti
secure-IV-TH-INF
‘be securing’ | b. osigur-av-a-telj
secure-IV-TH-N
‘insurer’ |

However, Slavic nouns have also been argued to have theme vowels (Halle 1994, Bailyn & Nevins 2008, Halle & Nevins 2009, a.o.) and Slavic adjectives have been claimed to share the theme vowels of nouns (Halle & Matushansky 2006).⁹ If all major word classes in BCS have themes (as proposed for Russian in Halle 1994), themes could then equally well be attributed to roots, as in (41), with contextual allomorphy able to work in the familiar way predicted by the cyclicity schema in (7) and the activity corollary in (8). For instance, the theme vowels in (42) could all be due the root $\sqrt{\text{ZEN}}$, the surface allomorph of the theme then being conditioned by the element that selects the root+theme combination (i.e., the first categorizer or another root).

- | | | |
|------|--|---|
| (41) | a. ROOT-TH- <i>n</i>
b. ROOT-TH- <i>v</i> | c. ROOT-TH- <i>a</i>
d. ROOT-TH-ROOT |
|------|--|---|

⁸Note that this does not force the conclusion that *all* (derivational) affixes are roots, as in the radical proposal of Lowenstamm 2010.

⁹As was the case with some agent nominals, themes are deleted on the surface if they are followed directly by a vowel-initial suffix. This is a regular hiatus resolution strategy in Slavic.

- (42) a. *žen-a-Ø-ma*
 woman-TH-N-DAT.SG
 ‘to the women’
 b. *žen-i-Ø-ti*
 woman-TH-V-INF
 ‘marry a woman’
 c. *žen-Ø-in*
 woman-TH-A.POSS
 ‘woman’s’
 d. *žen-o-mrz-Ø-ac*
 woman-TH-hate-TH-N
 ‘woman-hater’

The surface form of the theme may also be determined by the root it attaches to; this would need to be the case for AV, which always conditions the theme *-a*, regardless of the categorizer that follows. If this approach turns out to be correct, then the presence of a theme vowel in AV-containing agent nominals does not necessarily indicate the presence of a verbal categorizing morpheme. Instead, it provides further evidence that AV is a root, which conditions the appearance of a theme vowel like other roots.

4.3 Lexical prefixes

There is a well-established tradition in generative linguistics of treating so-called ‘lexical prefixes’ on verbs as proprietary to the verbal extended projection, where they are usually argued to contribute a result component (e.g., [Svenonius 2004c](#), [Arsenijević 2006](#), [Ramchand 2008](#), a.o.). In addition to having this aspectual component, lexical prefixes change the meaning of the root/stem they attach to in a way that is not predictable from the meaning of the prefix and the meaning of the root (43a-b).¹⁰ In (43c), I provide the agent nominal based on the stem *prouč-*; while the lexical meaning component of the prefix is retained, there is no discernible result aspect to this nominal.

- (43) a. *uč-i-ti*
 learn-TH-INF
 ‘be learning’
 b. *pro-uč-i-ti*
 LP-learn-TH-INF
 ‘examine (smth)’
 c. *pro-uč-av-a-telj*
 LP-learn-AV-TH-N
 ‘researcher’

There are two main questions that arise in the domain of lexical prefixes on nouns, in my view. First, is the prefix, if it is an independent morpheme, necessarily verbal? These prefixes are pervasive in the nominal domain, for example, (44). A verbal treatment of lexical prefixes would require us to treat all of those nouns as deverbal, with no other evidence that this is the case. This difficulty has already been discussed by [Svenonius \(2004b\)](#), who argues that the prefixes have a distinct origin on root-nominals and on verbs. Specifically, he argues that

¹⁰The prefix *pro-* has no discernible meaning of its own in the synchronic grammar of BCS.

“when a lexical prefix and a root are stored as an idiom, they are accessible for a compounding process” (Svenonius 2004b:241). This is precisely what I will argue is the case for BCS lexical prefixes in AV-containing agent nominals, see section 4.5. There is also some phonological support for the distinction between lexical prefixes on verbs and on root-nominals: Scheer 2001 shows that Czech nouns that have lexical prefixes systematically have long vowels on the prefix, which is not true of the corresponding verbs.¹¹ This suggests that lexical prefixes are perhaps added to nominal and verbal structures in distinct ways.¹² Something similar may be going on in BCS, as the vowel on the prefix is short on the verb in (44a), but long on the nouns in (44b) and (44c); however, note that the accent pattern of the two is also different, which may be driving the length difference. This issue clearly merits further research, which is beyond the scope of this paper.

- | | | | |
|------|-----------------|-------------------|-------------------|
| (44) | a. na-uč-í-ti | b. naa-úk-a | c. naa-úk-Ø |
| | LP-learn-TH-INF | LP-learn-N.NOM.SG | LP-learn-N.NOM.SG |
| | ‘learn’ | ‘science’ | ‘lesson’ |

Polish nouns of the type shown in (44) are argued to be deverbal in Łazarczyk 2010, based on two main arguments. The first argument invokes agent nominals of the kind discussed in this paper. Łazarczyk argues that the presence of the morpheme AV (-i/-ow) is required in these nominals because it acts as an atelicizer for prefixed stems “in order to compose with suffixes that require an activity-denoting (as opposed to an accomplishment- or achievement-denoting) stem” (Łazarczyk 2010:87). As we have seen for BCS, many prefixed stems do include the morpheme AV in agent nominals, but this preference is not absolute, recall (34). Moreover, we have seen agent nominals based on the same root with and without the morpheme AV, with no apparent difference in meaning (35). Therefore, this argument does not go through in the case of BCS agent nominals.

The second argument is that the prefixed nominals have “result readings”, illustrated with nouns like *przy-pis* ‘footnote’ from *przy-pisać* ‘to attribute’. However, that these nouns have “result readings” is simply stated, rather than argued for (also in Caha & Ziková 2022 for Czech). In terms of the Polish example above, the result of an attributing event is an attribution but not necessarily a footnote.

¹¹There are additional phonological constraints; for instance, the vowel on the prefix is long only if the noun stem has a short vowel.

¹²A reviewer asks how this paper accounts for the systematic aspectual effect of the lexical prefixes in verbal contexts if the prefixes are not themselves verbal. The view presented here, under which lexical prefixes are added to nominal and verbal structures in distinct ways, may provide a natural explanation within DM. Lexical prefixes may get aspectual interpretation in the context of verbal structure, but not otherwise.

As for the BCS cases mentioned above, is the result of the learning event denoted by (44a) supposed to be science (44b) or a lesson (44c)? It is not clear to me that we need to invoke the result component of the corresponding verbal predicate to arrive at the meanings of these nouns. At the very least, this would need to be argued for, especially given the general concerns about decomposition of stems and lexical prefixes I express immediately below.¹³

The second central question in the domain of lexical prefixes, which has received surprisingly little attention given the pervasive decomposition of lexically-prefixed stems in the literature, is the following: Is the composition is always or ever synchronic (even on verbs) and is it computed by the grammar? Analytical work has thus far not provided any convincing arguments for either of these positions and it is unclear to me what the shape of such an argument would even be. One alternative may be that prefixed and unprefixed verbs that (seemingly) contain the same stem belong to the same morpho-syntactic classes because of morpho-phonological similarity or because of diachronic relatedness. Therefore, the rest of this section offers some speculative remarks on how these issues could be explored using experimental methods.

Kazanina 2011 is the only study on a Slavic language to date which attempted to answer this question (though there are issues with the methodology; see below). **Kazanina** looked at the decomposition of prefixed words in Russian by doing a series of masked priming experiments, in which she found facilitation both when the prime and the target were (assumed to be) morphologically related (e.g., *rost* ‘growth’ vs. *na-rost* ‘outgrowth’) or only apparently morphologically related (e.g., *ton* ‘tone’ vs. *priton* ‘den’). She concludes that “all orthographic forms that can be exhaustively parsed into a prefix and a stem are decomposed into (apparent) constituent morphemes during their retrieval from the lexicon.”

However, it is important to emphasize that this does not provide evidence that lexical prefixes are decomposable in the grammar. This kind of automatic decomposition is unlikely to be driven by the grammar, a point that is frequently brought up for the comparable English paradigm. The paradigm in question includes truly suffixed words (*cleaner* → *CLEAN*), pseudo-suffixed words (*pigment* → *PIG*), and words that are only orthographically related (*brothel* → *BROTH*). In masked priming, facilitation is reliably observed in the pseudo-suffixed condition (**Rastle & Davis 2003**, **Marslen-Wilson, Bozic & Randall 2008**, **Beyersmann et al.**

¹³**Basilico (2008)** argues that the lexical prefixes do introduce a state, but that they merge with roots before the categorizing head, which is an analysis that incorporates the intuition that the nouns in question have a state component, but would be compatible with the proposal made in this paper, since the prefixes are not verbal in any way.

2016, a.m.o). However, this does not compel researchers to conclude that the English word *pigment* is morphologically decomposable; instead, the discussion has revolved around understanding the properties of early visual-word recognition that may be driving the effect. In turn, we should not conclude that Slavic lexical prefixes are decomposable on the basis of such experimental results either.

A different, potentially useful kind of priming study is one which has been used most extensively on Semitic languages, which have non-concatenative (templatic) morphological systems. The aim in these studies has been to understand whether roots—which never appear as a unit but are always found inside categorizing templates—are treated by the morphosyntax as separate from those templates (see the early work of Frost et al. 1997; Berrebi et al. 2023 for an overview). The single most robust experimental result in this literature is what is referred to as ‘root priming’: When a prime and a target share the same root, participants are faster at identifying the target as a real word compared to a control, where the overlap in form is purely orthographic.

To exemplify how this would work in the case of BCS prefixes, if the forms in (45) share a root, we would expect faster word recognition in (45a-b) than in (46a-b), which do not share a root but whose forms overlap in the same number of elements. There are additional methodological considerations, for example the use of auditory (unmasked) primes in a continuous lexical decision task as ways to minimize the role of visual processing, but I do not explore them in detail here (see e.g., Creemers et al. 2023 for discussion).

- | | | |
|------|------------------------------------|---|
| (45) | a. d-a-ti
give-TH-INF
‘give’ | b. pro-d-a-ti
LP-give-TH-INF
‘sell’ |
| (46) | a. vin-a
wine-NOM.PL
‘wines’ | b. osovin-a
axis-NOM.SG
‘axis’ |

To conclude the discussion of lexical prefixes, let me spell out what should be obvious from some of the examples seen throughout this paper: Nouns with lexical prefixes do not behave like re-categorized (deverbal) nouns for purposes of the morpho-phonology. Root-conditioned allomorphy and accent realization are available to *n* freely in nouns that contain lexical prefixes, suggesting that these prefixes are not verbal. Further (likely experimental) evidence is needed to determine whether lexical prefixes are decomposable in the synchronic grammar in the first place.

4.4 Syntactic evidence?

I only briefly note here that there is no syntactic (i.e., distributional) evidence that AV-containing agent nominals have verbal structure. For example, if AV corresponds to the (grammatical) Asp(ect) head, we may expect agent nominals that contain it to be able to have accusative complements like their corresponding verbs, contrary to fact (47a). In reality, agent nominals in BCS can only have genitive-marked arguments, which is true of BCS nouns more generally (47b).

The accusative case expectation comes from two widespread assumptions: (i) that the domain of grammatical aspect is above the argument structure domain, and (ii) that the argument structure domain includes VoiceP, which is associated with accusative-case licensing. In a monotonic fashion, we then expect structures that include Asp to also include Voice (48).¹⁴

- (47) a. prod-av-a-ac (cipel-a / *cipel-e)
 sell-AV-TH-N shoe-GEN.PL shoe-ACC.PL
 ‘shoe seller’

- b. ugao (ulic-e / *ulic-u)
 corner street-GEN.SG street-ACC.SG
 ‘street corner’

- (48) [... Asp [... Voice ...]]

If AV does not correspond to Asp but to another head in the verbal extended projection, then the expectation would be different. For example, I have noted that a number of authors take AV to be very low in the verbal spine and contribute an atelicizing or eventizing meaning component. However, as I have noted, there seems to be no correlation between the presence/absence of AV and the interpretation of the agent as an agent of an atelic or telic event.

It is unclear what the expectation is if the agent nominals I discuss contain only *v* and some very low, non-categorizing verbal prefixes which have no clear effects on the distribution of the expression they are found in. If these nominals contain *v*, we may expect this to correlate with the obligatory presence of a (genitive-marked) internal argument, in cases where the corresponding verbs are transitive. However, as seen in (47a), the internal argument is optional. I therefore conclude that there is no compelling distributional evidence that these agent nominals contain any verbal structure.

¹⁴I refer the reader to Baker & Vinokurova’s (2009) typological study, which finds that no agent nominals in the 78 languages they examine can take an accusative object.

4.5 The structure of AV-containing agent nominals

I have argued that AV-containing agent nominals are root-root compounds. In this section, I examine the structural properties of these elements in more detail.¹⁵ In section 3.2., I proposed that certain BCS elements, which uncontroversially contain two roots, should be analyzed as root-root compounds (49). They frequently involve bound roots (which cannot appear as words independently), they may include a Linker, and they contain only one accent, which is assigned in the domain of the first categorizer in BCS. We contrasted these with compounds that involve two categorized elements, are not connected via a Linker and each element has its own accent (50). I then argued that AV-containing agent nominals pattern with root-root compounds in a number of ways.

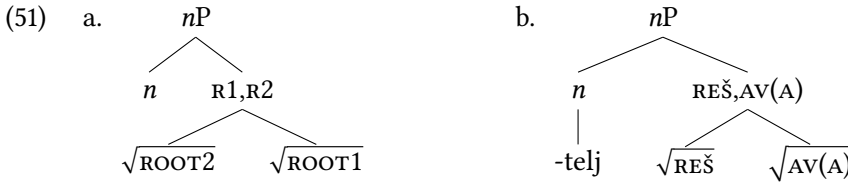
- | | | |
|------|--|--|
| (49) | a. dub-o-rez-ác
deep-L-cut-N
‘woodcarver’ | b. pad-o-bran-ác
fall-L-defend-N
‘parachuter’ |
| (50) | a. nadžák bába
pickaxe.N grandma.N
‘belligerent person’ | b. rák rána
cancer.N wound.N
‘sore subject’ |

Root-root compounds have not received a great deal of attention in the formal morphosyntactic literature. One of the foundational tenets of DM is that roots are acategorical, that is, syntactic terminals which do not have a category. However, roots cannot appear bare; they must merge with categorizing heads in order to be interpretable at the interfaces (Marantz 1997, 2000, 2001, Embick & Marantz 2008). In practice, this has meant that roots are standardly assumed to merge with categorizers before they merge with any further functional structure.¹⁶ However, elements similar in a number of respects to the BCS ones in (49) have been noted to exist in a number of languages and argued to be root-root compounds (see Zhang 2007 for Chinese, Borer 2013 for English, Newell & Piggott 2014 for Ojibwe, De Belder 2017 for Dutch).

¹⁵Note, however, that the proposal in this section is orthogonal to the arguments made in the rest of the paper. That is, even if the structure of root-root compounds proposed here turns out to be wrong, this would have no direct bearing on the arguments we have seen regarding the lack of verbal structure in AV-containing agent nominals.

¹⁶Marantz (2001) explicitly departs from this assumption, arguing that affixes can be attached to roots prior to categorization. If this is the case, then (i) AV (and other such elements) may not be roots per se; their properties may follow from merging with the root prior to categorization and (ii) we may then need to seek alternative evidence to determine their identity.

I therefore assume with these authors that roots can be merged directly, prior to categorization. Hence, the structure of root-root nominal compounds is abstractly as in (51a), and illustrated for AV-containing agent nominals in (51b); see Bešlin 2025a for arguments that the identity of both roots must be visible to the first categorizing head. Recall that it is not clear whether lexical prefixes are distinct morphemes in the synchronic grammar of BCS. If it turns out that they are not, then they are simply part of the (exponent of the) root they precede. If they are, in fact, distinct morphemes, then they may also be merged into the structure prior to categorization. If merger prior to categorization is possible, there is in principle no reason why it should be restricted to two elements.¹⁷



There are a number of further issues regarding the structure of root-root compounds that I will leave open here. For example, I do not take a stance on whether roots in root-root compounds are always merged in binary structures, or whether ternary (n -ary) branching is also possible. Furthermore, I do not engage with the question of whether the *l*(inker) and theme vowels are present in the syntax of root-root compounds or whether they are dissociated morphemes (as proposed for Romance theme vowels in Oltra-Massuet 1999). What is crucial is only that these roots are combined before any categorization takes place.

5 Event/agent entailments $\neq$ verbal syntactic structure

In this section, I address the worry that the agent nominals discussed in this paper may not be root-derived because they trigger event and/or agent entailments. I argue against a strand of research that associates the existence of event and/or agent entailments with the obligatory presence of verbal syntactic structure.

In earlier generative work on this topic, eventive (episodic) interpretations with agent nominals were taken to correlate perfectly with the presence of complement structure (see Alexiadou 2001). For example, *frequent* in (52a) was argued to trigger an episodic interpretation of the agent nominal, in turn making

¹⁷Though see De Belder 2017 for some theory-internal objections and a proposal to derive the cluster of roots at PF via fission.

the complement obligatory to the same extent as for the corresponding verb *consume*. These kinds of examples were convincing because they seemed to involve both syntactic and semantic evidence for the presence of verbal structure. However, it is not difficult to come up with examples where episodic interpretations do not correspond to obligatory complement structure (52b); contrast this with *She frequently visited *(this region)*. Moreover, the noun in (52c) has an obligatory complement despite not being derived from an existing verb. This immediately puts into question the solidity of the original argument that these are valid diagnostics for the presence of verbal structure.

- (52) a. a frequent consumer *(of tobacco)
 b. a frequent visitor (of this region)
 c. a frequent subject *(of Monet's paintings)

More radically, Alexiadou & Schäfer 2010 extend the deverbal analysis to dispositional agent nominals, which are not eventive; they denote entities which are designated for some specific job or function but which do not have to be actually been involved in such a job or function. Some examples are given in (53). Alexiadou & Schäfer argue that even dispositional nominals have an articulated verbal structure, including *v*, Voice, and Asp.

- (53) a. fire-fighter
 b. teacher
 c. basketball player

This analysis raises a series of questions. Firstly, if these nominals contain rich verbal structure, why do they never have accusative-marked complements or adverbial modifiers (like gerunds)? The analysis is then completely divorced from any syntactic facts and is meant to account for the agent and/or event entailment patterns associated with these nominals. Relying on entailments to diagnose syntactic structure is a slippery slope, however. To give a couple of specific examples, it has sometimes been argued that the ambiguity of (54a), in particular the reading which allows the dancing to be characterized as beautiful, suggests that *dancer* is derived from the corresponding verb *dance*. Note, however, that the same ambiguity obtains in (54b-c), yet those nouns have no corresponding verbs to be derived from.

- (54) a. a beautiful dancer
 b. a beautiful violinist

c. an elegant midfielder

A similar point pertains to (55): One can be *a just ruler* by virtue of being just in their ruling capacity (55a), where *just* then seemingly modifies the event of ruling. However, if this is taken as evidence that *ruler* contains verbal structure, then the same must be said for *king* in (55b): One can be just in their capacity as king without necessarily being a just person. While I am not saying that analyzing *king* as a deverbal noun is obviously wrong, it is important to point out the logical endpoint of taking entailments as evidence for syntactic structure, especially in cases where such an analysis is likely less convincing for most.

- (55) a. a just ruler
b. a just king

All of this is, of course, part of a broader point: Entailments do not necessarily reveal the presence of hidden structure. Take (56) and the various entailments it licenses. For example, the truth of the statement that the child is blond can only be evaluated in virtue of the child having hair. However, this does not compel us to claim that (56) contains a hidden ‘hair’ argument that projects syntactic structure. A similar point can be made for the modifier *illegitimate* in (56). The agent nominal cases should be no different in this respect, as far as I can see.

- (56) an illegitimate blond child

The takeaway from this discussion is the following: In the absence of clear syntactic evidence, we should consider more carefully what kinds of semantic arguments are valid for diagnosing the presence of syntactic structure. For example, while scope facts are a reliable semantic cue for the presence of syntactic asymmetries, entailment patterns are less convincingly so (see Williams 2015 for an extended argument to this effect). Hence, I do not take agent/event entailments as knock-down arguments that the agent nominals considered in this paper must contain verbal structure, especially since there is clear morpho-phonological evidence to the contrary.

6 Conclusion

Based on morpho-phonological evidence from BCS agent nominals, I have argued that elements which are traditionally analyzed as verbal in Slavic (‘verbal’ theme vowels, secondary imperfectivizers, lexical prefixes) are not verbal at all.

Illustrating with data from root-derived versus deadjectival agent nominals, I first showed that root-conditioned allomorphy and accent placement in BCS are limited to the first spell-out domain. The first spell-out domain may include multiple morphemes (roots, non-categorizing morphemes), but only one categorizer.

I then showed that ‘deverbal’ agentive nouns containing morphology analyzed as verbal behave for these morpho-phonological processes like root-derived nouns (or root-root compounds). Moreover, in addition to there being no syntactic evidence for verbal structure in these agentive nouns, I argued that event/agent entailments do not provide evidence for such structure either. Finally, I provided an alternative analysis for the identity of the ‘verbal’ morphemes in question. I argued that (a) the ‘secondary imperfectivizer’ AV should instead be analyzed as a root, (b) theme vowels are morphemes which attach to roots in Slavic more generally and do not signal the presence of any specific categorial structure, and (c) there are serious reasons to doubt that ‘lexical prefixes’ are part of the verbal extended projection, and experimental work is needed to determine if they are even synchronically separable from the roots they attach to.

Abbreviations

3	third person	N	noun
ACC	accusative	NEG	negation
A	adjective	NEUT	neuter
DAT	dative	NOM	nominative
DIM	diminutive	POSS	possessive
GEN	genitive	PTCP	participle
INF	infinitive	PL	plural
L	linker	SG	singular
LP	lexical prefix	SUP	superlative
M	masculine	TH	theme vowel

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